

INSTRUCTIONS

This exam paper has a total of 3 questions. You must answer **all 3 questions**. The total number of possible marks for this exam is 100. Each question can be addressed independently. It is recommended that you use the perusal time to figure out the order you will solve each question.

All answers need to be justified. Explanations of the steps used are essential, and will be carefully accounted for in assessment. Any assumption made should be consistent with the question, clearly stated and justified. Best marks are awarded to numerically correct answers with full worked solutions, but partial credits may be given to partially complete solutions.

QUESTION 1

Provide a **short response** to the following questions (a) - (h). You may include supporting diagrams, mathematics and examples (real or theoretical) in your responses.

- How do you determine the stability of a nonlinear system?
- Describe how sampling time T can affect tracking effectiveness?
- Consider a state space model with states $x(kT)$. Consider a transformed state space model with states $z(kT) = P^{-1}x(kT)$ where P is a transformation matrix. Do these systems have the same stability properties? Justify your answer.
- Consider the state space model and pulse transfer function given in (1). Do the state space model and pulse transfer function represent the same system? Justify your answer.

$$x(kT + T) = \begin{bmatrix} 0 & -2 \\ 7 & 3 \end{bmatrix} x(kT) + \begin{bmatrix} \beta \\ 1 \end{bmatrix} u(kT), \quad F(z) = \frac{2z}{z^2 - 4z + 16} \quad (1)$$

- Describe the difference between state feedback control with integral action and state feedback control with feedforward gain with respect to control capabilities.
- State feedback control design allows the designer to place closed loop system poles anywhere in the s or z plane. Do optimal control approaches (i.e. LQR) allow the designer to place closed loop system poles anywhere in the s or z plane? Justify your answer.
- What is the purpose of the matrix R used in optimal observer (Kalman filter) design with respect to expected observer performance?
- Regarding observer design, consider placing the poles of the estimation error dynamics at either the points ($\times$) in diagram (i), (ii) or (iii) shown in Figure 1. Which of these three diagrams provides a better design for a continuous time observer? Justify your answer.

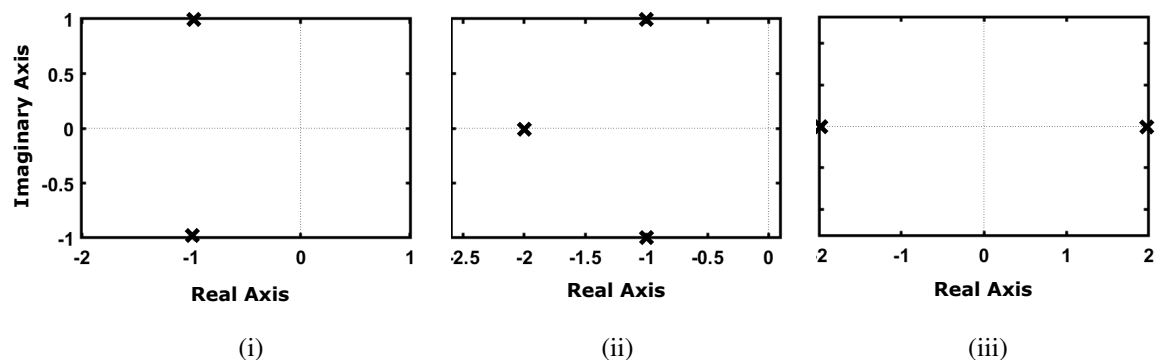


Figure 1: Candidate desired poles ($\times$).

(30 marks)

QUESTION 2

Space vehicles often carry sensing arrays. These arrays are directed at different stellar objects during a mission so precise attitude (orientation) control is required. Rotation of the platform about each axis is regulated by separate control systems of the same type. The state space model for this type of control system is given by (2) below.

$$\begin{aligned} \begin{bmatrix} x_1(kT + T) \\ x_2(kT + T) \end{bmatrix} &= \begin{bmatrix} 0.00 & -0.16 \\ 1.00 & -1.00 \end{bmatrix} \begin{bmatrix} x_1(kT) \\ x_2(kT) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(kT) \\ y(kT) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(kT) \\ x_2(kT) \end{bmatrix} \end{aligned} \quad (2)$$

where x_1 is angular position, x_2 angular velocity, u is the input control and y is the measured output. An space technology startup wishes to use this model to design digital attitude control systems for various satellites and missions.

- (a) Design a digital full-order state observer to estimate the system states where the equivalent continuous time observer poles are located at $p_1 = -10$ and $p_2 = -20$. State your assumptions.
- (b) Write an expression for the discrete time state space model for the closed loop system with state feedback control and full order state observer. You may assume the gain K and L for the controller and observer receptively.

(30 marks)

QUESTION 3

The longitudinal (forward-backward) motion of an autonomous car depends on its physical characteristics and environmental conditions (road and weather). A simplified incremental linearised discrete time model representing the longitudinal motion is given by (3) below

$$\begin{bmatrix} v(kT + T) \\ q(kT + T) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v(kT) \\ q(kT) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} (\theta(kT) - b(kT)), \quad y(kT) = \begin{bmatrix} 0 & \alpha \end{bmatrix} \begin{bmatrix} v(kT) \\ q(kT) \end{bmatrix} \quad (3)$$

where v is the velocity, q is the acceleration, θ is the throttle opening, b is a transformed brake pressure, y is the measured output and α is a constant. An automotive company wishes to use this model to design digital cruise control systems for various car models and road conditions.

- (a) Provide an analysis of the model (3) that would help the company design digital cruise control systems that regulate velocity assuming a sampling time $T \leq 1$ second.
- (b) The company designs a digital state feedback controller to regulate velocity for a particular car model given by (4). Determine the expected performance of the controller, justifying your response with evidence.

$$\theta(kT) - b(kT) = -0.51 v(kT) + 0.43 q(kT) \quad (4)$$

- (c) The company wants to re-design the digital state feedback controller in part (b) such that the response balances control effort and state deviations. Design an optimal state-feedback controller to achieve this objective using the following matrices (5).

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = 1, \quad P = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \quad (5)$$

Hint: You must show all design steps, but do NOT need to solve the Riccati equation to find P as an intermediate step.

- (d) Highlight some practical issues or considerations when implementing the controllers designed in part (b) and (c). Can you suggest an alternate modelling or design approach?

(40 marks)

END OF PAPER

(i)

RESOURCE SHEET

	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$
1.	—	—	Kronecker delta $\delta_0(k)$ 1 $k = 0$ 0 $k \neq 0$	1
2.	—	—	$\delta_0(n-k)$ 1 $n = k$ 0 $n \neq k$	z^{-k}
3.	$\frac{1}{s}$	$1(t)$	$1(k)$	$\frac{1}{1-z^{-1}}$
4.	$\frac{1}{s+a}$	e^{-at}	e^{-akT}	$\frac{1}{1-e^{-aT}z^{-1}}$
5.	$\frac{1}{s^2}$	t	kT	$\frac{Tz^{-1}}{(1-z^{-1})^2}$
6.	$\frac{2}{s^3}$	t^2	$(kT)^2$	$\frac{T^2 z^{-1}(1+z^{-1})}{(1-z^{-1})^3}$
7.	$\frac{6}{s^4}$	t^3	$(kT)^3$	$\frac{T^3 z^{-1}(1+4z^{-1}+z^{-2})}{(1-z^{-1})^4}$
8.	$\frac{a}{s(s+a)}$	$1 - e^{-at}$	$1 - e^{-akT}$	$\frac{(1-e^{-aT})z^{-1}}{(1-z^{-1})(1-e^{-aT}z^{-1})}$
9.	$\frac{b-a}{(s+a)(s+b)}$	$e^{-at} - e^{-bt}$	$e^{-akT} - e^{-bkT}$	$\frac{(e^{-aT} - e^{-bT})z^{-1}}{(1-e^{-aT}z^{-1})(1-e^{-bT}z^{-1})}$
10.	$\frac{1}{(s+a)^2}$	te^{-at}	kTe^{-akT}	$\frac{Te^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$
11.	$\frac{s}{(s+a)^2}$	$(1-at)e^{-at}$	$(1-akT)e^{-akT}$	$\frac{1-(1+aT)e^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$
12.	$\frac{2}{(s+a)^3}$	$t^2 e^{-at}$	$(kT)^2 e^{-akT}$	$\frac{T^2 e^{-aT}(1+e^{-aT}z^{-1})z^{-1}}{(1-e^{-aT}z^{-1})^3}$
13.	$\frac{a^2}{s^2(s+a)}$	$at - 1 + e^{-at}$	$akT - 1 + e^{-akT}$	$\frac{[(aT-1+e^{-aT}) + (1-e^{-aT}-aTe^{-aT})z^{-1}]z^{-1}}{(1-z^{-1})^2(1-e^{-aT}z^{-1})}$
14.	$\frac{\omega}{s^2 + \omega^2}$	$\sin \omega t$	$\sin \omega kT$	$\frac{z^{-1} \sin \omega T}{1-2z^{-1} \cos \omega T + z^{-2}}$
15.	$\frac{s}{s^2 + \omega^2}$	$\cos \omega t$	$\cos \omega kT$	$\frac{1-z^{-1} \cos \omega T}{1-2z^{-1} \cos \omega T + z^{-2}}$
16.	$\frac{\omega}{(s+a)^2 + \omega^2}$	$e^{-at} \sin \omega t$	$e^{-akT} \sin \omega kT$	$\frac{e^{-aT} z^{-1} \sin \omega T}{1-2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$
17.	$\frac{s+a}{(s+a)^2 + \omega^2}$	$e^{-at} \cos \omega t$	$e^{-akT} \cos \omega kT$	$\frac{1-e^{-aT} z^{-1} \cos \omega T}{1-2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$
18.	—	—	a^k	$\frac{1}{1-az^{-1}}$
19.	—	—	a^{k-1} $k = 1, 2, 3, \dots$	$\frac{z^{-1}}{1-az^{-1}}$
20.	—	—	ka^{k-1}	$\frac{z^{-1}}{(1-az^{-1})^2}$
21.	—	—	$k^2 a^{k-1}$	$\frac{z^{-1}(1+az^{-1})}{(1-az^{-1})^3}$
22.	—	—	$k^3 a^{k-1}$	$\frac{z^{-1}(1+4az^{-1}+a^2 z^{-2})}{(1-az^{-1})^4}$
23.	—	—	$k^4 a^{k-1}$	$\frac{z^{-1}(1+11az^{-1}+11a^2 z^{-2}+a^3 z^{-3})}{(1-az^{-1})^5}$
24.	—	—	$a^k \cos k\pi$	$\frac{1}{1+az^{-1}}$

$x(t) = 0$ for $t < 0$
 $x(kT) = x(k) = 0$ for $k < 0$
 Unless otherwise noted, $k = 0, 1, 2, 3, \dots$